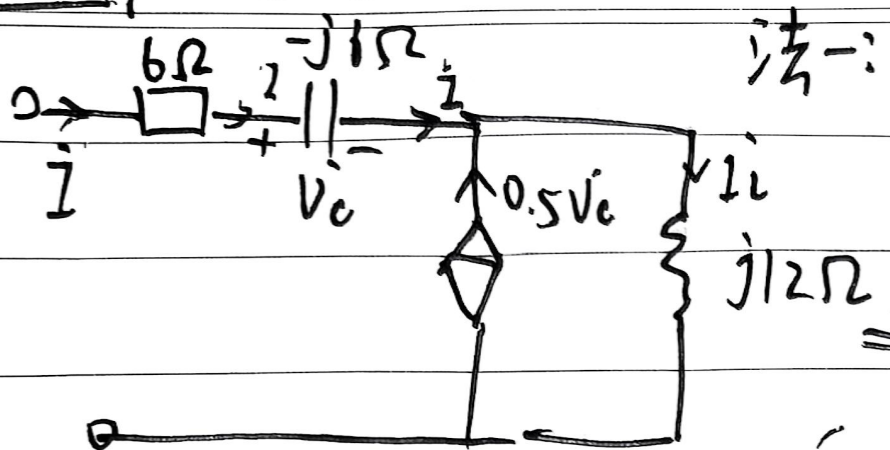


5.14



$$\begin{aligned} \dot{U} &= \dot{i} \cdot 6 + \dot{V}_c + (\dot{i} + 0.5\dot{V}_c) \cdot j12\Omega \\ \dot{V}_c &= \dot{i}(-j6\Omega) \\ \Rightarrow \dot{U} &= 4\dot{i} + j6\dot{i} \\ \therefore Z &= \frac{\dot{U}}{\dot{i}} = 4 + j6\Omega \end{aligned}$$

5.15

$$\dot{U}_N = -\dot{U}_b + \dot{U}_a$$

$$\dot{U}_a = 10 \angle 1000t + \frac{\pi}{3}$$

$$\dot{U}_b = \frac{1}{j\omega C} \dot{I}$$

$$\dot{U}_b = 5 \angle 1000t - \frac{\pi}{6}$$

$$\dot{U}_a = -\dot{I}R + \dot{U}_b$$

$$\therefore \dot{U}_N = \dot{U}_a - \dot{U}_b = 10 \angle 1000t + \frac{\pi}{3} - 5 \angle 1000t - \frac{\pi}{6} = 5 \angle 1000t + 1.511$$

$$\dot{I} = -j\omega C \dot{U}_b = 50j$$

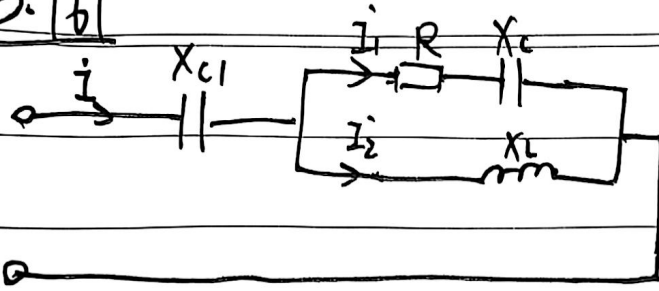
$$\Rightarrow Z = \frac{\dot{U}_N}{\dot{I}} = \left(\frac{\dot{U}_a}{\dot{U}_b} - 1 \right) (-j\omega C) = \left(2 \angle \frac{\pi}{2} - 1 \right) (-10j) = 20 + 10j$$

$$X = 10 = \omega L \Rightarrow L = 1 \times 10^{-2} \text{ H} ; R = 20 \Omega$$

$$Y = \frac{1}{Z} = \frac{20 - 10j}{500} = 0.04 - j0.02 \text{ S} = G - jB$$

$$R' = \frac{1}{G} = 25 \Omega, B = 0.02 = \frac{1}{\omega L} \Rightarrow L = 5 \times 10^{-2} \text{ H}$$

5.16



$$\omega = 100 \text{ rad/s}$$

$$\dot{I} = 5 \angle \omega t + \frac{\pi}{4}$$

$$\dot{U} = 50\sqrt{2} \angle \omega t$$

$$\begin{cases} \dot{U} = j\dot{I}X_{c1} + j\dot{I}_2X_L \\ \dot{I}_2X_L = \dot{I}_1(R + jX_c) \\ \dot{I} = \dot{I}_1 + \dot{I}_2 \end{cases} \Rightarrow \dot{U} = j\dot{I}(X_{c1} + \frac{R + jX_c}{1 + j\frac{R + jX_c}{X_L}})$$

$$\Rightarrow 10j + \frac{R + jX_c}{1 + j\frac{R + jX_c}{X_L}} = \frac{\dot{U}}{\dot{I}} = 10\sqrt{2} \angle -\frac{\pi}{4} = 10 - 10j$$

$$\Rightarrow R + jX_c = 5 - 5j$$

$$\Rightarrow \begin{cases} R = 5 \Omega \\ X_c = 5 \Omega \end{cases}$$